Vibe Coding For AI Application Development

Teaching Record

24-07-2026

Scope

- Vibe Coding
  - What/Why/How
  - Tools
  - Python
  - IT Essentials
- AI Application Development
  - Workflow
  - Chatbot
  - Agents

Scope

- Overview of AI Applications with Vibe Coding
- Setup of tools
- Python with vibe coding
- Claude Code
- Use cases of AI applications
- Agentic use cases with vibe coding

Use cases of AI Solution Development

- Document Summarization Engine
  - OCR detection engine
- Database Design and deployment
  - Text2SQL
- Web scraping techniques
  - News sentiment Analysis engine
- Multi-Agents deployment

Class One

- Overview of LLM Solutions
  - Prompts
  - Coding LLM
  - Agent
    - What is Agent
      - Execution
      - LLM (Brain)
      - Tools (Execution)
      - Memory
      - Harness (looping and orchestration)
- Setup of tools
  - Cursor
    - Subscription
  - Grok/Gemini (Browser mode LLM)
  - Claude code CLI (To be continued)
    - Cursor
    - Openrouter (LLM)
- Use sample py program to read it with coding agent

Class one

- Use case of using cursor
  - Read existing python script
  - Execute the python script
- Prompt techniques (To be continued)
- Python programming with vibe coding

Class two

- Recap of last lesson
  - **Grok (Browser)**
  - **Cursor**
    - **Write the program(s) in the working folder**
    - **Debug the program(s) in the working folder**
    - **Deploy the solution in cloud**
  - Claude code
- Prompt techniques
  - System prompt
  - User prompt
  - Zero shot
  - Few shot
- Python programming with vibe coding
  - Tools (Anaconda) (Jupyter notebook)
  - Data types
  - Function
  - **Data structures (To be continued)**
- **Class exercise**
  - **Vibe coding**
    - **Generate stock processing excel**

Class two

- Function
  - From python native functions
    - print()
  - From libraries
    - yfinance
      - download()
    - yf.download()
  - Define custom function
- Packages
  - pip install
- Python
  - Data types

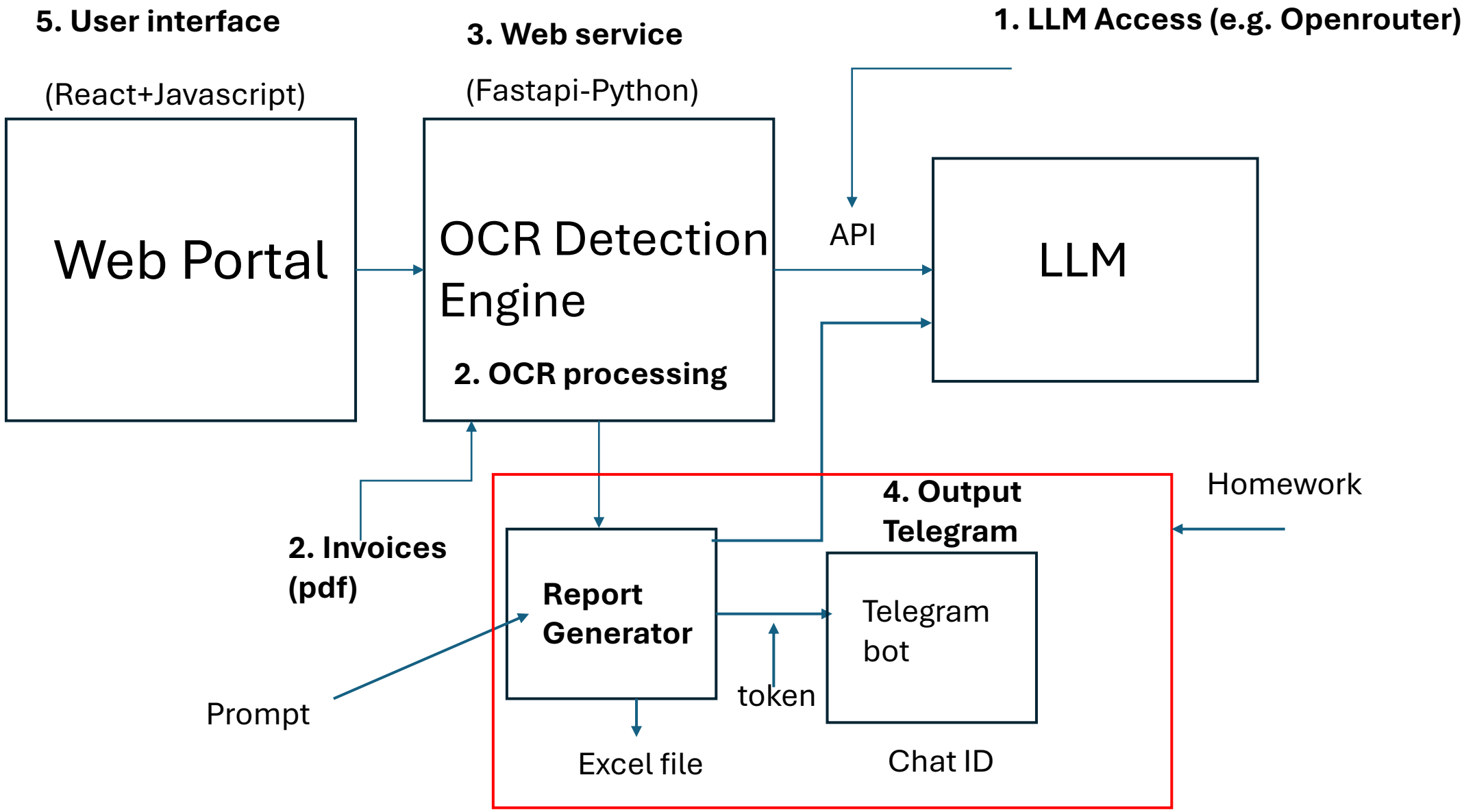
Class three

- Data objects
  - List
    - append()
  - Dictionary
  - Pandas
    - Series
    - Dataframe
      - describe()
      - Add column
      - Filtering by position
- Homework

Class four

- LLM API
- OCR Detection engine (Base module)
- Class exercises (Vibe Coding)
- Task automation with Python
- Development of SDD
- Development of OCR detection engine (Front end + Backend)
- Telegram API
- Class exercises

Home work


5. User interface
(React+Javascript)
3. Web service
(Fastapi-Python)
1. LLM Access (e.g. Openrouter)
Web Portal
OCR Detection Engine
2. OCR processing
API
LLM
2. Invoices (pdf)
4. Output Telegram
Homework
Report Generator
Telegram bot
Prompt
token
Excel file
Chat ID

Class five

- Homework
- Telegram API
- n8n
- MCP
- Fastapi (Python) (To be continued)

Homework one

- Using the OCR module, add the telegram module and send out message.
- In the front end , when user click the button, it will send the excel file via telegram

Homework two

- Modify the existing n8n workflow (Personal assistant v3 (6-Aug-2026).json) with the following features:
- UI: Telegram trigger
- AI Agent: Revise the system prompt
- LLM: via openrouter

- Use case: This is a standard AI chatbot and it will reply to user’s question.

Class six

- Setup of VSC+Github copilot+Openrouter API
- Database setup
  - Relational DB
    - Class exercise
    - Home work (text2SQL)
  - Vector DB
    - RAG (Document searching engine)
- Setup public URL with ngrok
- AI Chatbot
- Agentic assistant (n8n)+**Web services with Python (Fastapi) (To be continued)**

Steps to get public URL and work with n8n

- Login ngrok
- Copy key and perform authentication
- Setup public URL and port forwarding
- Setup environment variable to indicate the public URL for webhooking

Homework
Text2SQL

